

Analyzing the Extent to which the Preservation
Experience of the Nord-Pas de Calais Mining Basin is
Applicable to the Fushun West Open-Pit Mine's
Preservation via Les Argales Park and Le Métaphone

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1. Introduction

It is Fushun Social and Economic Development Research 2018 (Fushun Bureau Group of Social Science 2018) that proposed the idea that the Fushun West Open-pit mine can draw experience from the Nord-Pas de Calais (NPC) Mining Basin, which is described as a “successful and prestigious preservation model” since “it was inscribed in the UNESCO World Heritage List in 2012.” This proposal implies that Fushun's local planners view the NPC Mining Basin as a template for preservation in the West Open-pit mine.

However, Fushun Social and Economic Development Research 2018 merely mentioned this idea of learning from the Nord-Pas de Calais Mining Basin. It gave two short reasons: “similar historical background” and “similar architecture.” However, it didn’t provide details to support them. There remains a research gap in studying the actual applicability of foreign preservation experience. Therefore, this essay will examine the extent to which French preservation experience can be applied to the Fushun West Open-pit mine via two specific programs in the NPC Mining Basin – Le Métaphone and Les Argales Park.

2. Historical Context

The NPC Mining Basin consists of coal mines. According to an introductory document produced by the local preservation committee and UNESCO that details the NPC Mining Basin’s history, the NPC Mining Basin is located on “a coal seam.” It states that “the first mining company was founded in 1757” and “the last coal” stopped producing “on 21 December 1990.” This document provides primary historical evidence for its claim.

Victor Seow is an associate professor of the social science at Harvard University. His publication on China’s industrialization directly details the history of West Open-pit mine (Seow 2022). It claims that West Open-pit mine contained both coal and shale oil resources and operate from the 1900s to the 2000s.

The West Open-pit mine closes far later than the NPC Mining Basin, and this seems to partially explain why the NPC Mining Basin starts preservation earlier and achieve more results. Admittedly, a full explanation requires additional considerations.

3. Geological Context

Land subsidence means that surface settles downward when the materials below can no longer support it.

Subsidence happens in both the NPC Mining Basin and West Open-pit mine. Both Wu et al. and Guéguen et al. attribute subsidence to the removal of lower mined layer because they fail to support upper ground layer (Wu et al. 2025; Guéguen et al. 2009).

Guéguen et al.'s research in "Journal of Applied Geophysics" conducts valid measurements of subsidence levels in the NPC Mining Basin caused by mining exploitation and briefly compare it with that of West Open-pit mine (Guéguen et al. 2009). The main content of their research is to track the subsidence level over a 15-year period (1992-2007) through two new advanced technologies claimed to be more accurate. They find that the subsidence level is around 1cm / year in the NPC Mining Basin. This level seems far lower than that of West Open-pit mine, whose subsidence level is studied by Wu et al. Wu et al.'s research identifies a rate far greater than 1cm / year on average.

This difference matters and probably makes many methods applicable in the NPC area less applicable to the West Open-pit mine. Subsidence is dangerous. Feng et al.'s case study of Fushun West Open-pit mine investigates the floods, landslides, and passive migration caused by the subsidence. Their research clearly shows that this subsidence causes substantial damage in Fushun city. Nevertheless, they seem to omit other factors that probably matter such as rainfall.

4. Evaluative Criteria

How to evaluate a method applied in heritage preservation?

Regarding heritage sites' preservation, Vafaie, Remøy, and Gruis's research builds an evaluative model that includes "ten categories of success factors." They are "architectural, structural, socio-cultural, economic, environmental, energy, authenticity, legal, management and functional factors." This model provides a with a framework for this essay to judge a specific method's applicability to the West Open-pit mine (Vafaie, Remøy, and Gruis 2023).

5. Les Argales Park

Les Argales Park involves two sections: an artificial lake (shown in the red circle) in the mine pit and a green walking path surrounding the lake and crossing "mine mountains" made of mining waste (shown in the black circle). There appear to be three methods represented by these two sections: water filling, vegetation planting, and building walking paths.



Figure 5.1 | Map of Les Argales Park

(Retrieved from https://www.tripadvisor.com/Attraction_Review-g4926630-d4922108-Reviews-Espace_Terril_des_Argales-Rieulay_Nord_Hauts_de_France.html)

5.1. Artificial Lake

First and foremost, artificial lake is a typical “static body of water.”

Wu et al.’s article in “Hydrogeology & Engineering Geology” titled “Instability Mechanism of Water-Wading Soft Rock Slope During Storage Period in Fushun West Open-Pit Mine” conducted a simulated experiment on the effect of static water by building a simulation platform with multiple parameters and clear margins of error (Wu et al. 2025). Though the simulation appears to be well-designed, the lack of field data will still weaken its credibility.

The experimental results show that even a low depth of static water would cause serious “hydrostatic uplift,” which means the soil layer surrounding the water cracks (Wu et al. 2025). This suggests that the proposed lake-building strategy would actively trigger further landslides in Fushun’s specific mudstone conditions. This would supposedly lead to floods, subsidence, and landslides. The current Fushun city,

according to Feng et al., who collected and analyzed past casualties and locations of geological disasters in the West Open-pit mine, is sensitive and vulnerable to these phenomena because Fushun's living areas and traffic routes are too close to the verge of the West Open-pit mine (Feng et al. 2024).

This potential danger makes this method appear to score particularly low in the categories of structural and environmental factors. This is because it is not compatible with existing the geological structure and likely to cause serious damage.

5.2. Green Walking Path

First, vegetation planting appears to prevent soil loss and restore polluted land. According to Xiao et al.'s article in "Advances in Engineering Technology Research", planting is a practical method of restoration (Xiao et al. 2024). The mechanism here is direct: plant roots grow deep into the soil. Altogether, these roots can stabilize the soil. However, this stabilizing effect is significant in large tree species and minor in small grass plants (Xiao et al. 2024). This biological intervention provides a natural method to mitigate soil erosion across the pit's exposed slopes. Nevertheless, Xiao et al. seem to omit the trade-off of this method; their article does not mention the possible cost and exact extent of this effect. Without knowing the cost, it is difficult to say whether this method is efficient.

Second, walking paths provide places to walk. Walking appears to help individuals recover from post-disaster trauma and recollect collective memory. This is implied by Murakami's personal narrative, "A Walk to Kobe." The narrator describes his feelings while walking in Kobe after an earthquake. This appears to support the recovery effect of walking. This stimulus material provides a key insight: walking paths may serve as a medium for recalling individual memories and recovering from a sense of lost.

There appears to be a collective disaster associated with the closure of the West Open-pit Mine and Fushun's broader industrial decline. According to Chen et al.'s article in "China Population Science" about the 1990s wave of unemployment that hit northern China including Fushun, the transformation from a planned economy to a market economy caused the closure of large state-owned enterprises that were in charge of major factories such as the West Open-pit Mine and many heavy industrial enterprises in Northeast China (Chen, Hu, and Fu 2017). This wave, accordingly, caused workers to lose their jobs. Conducting statistical tests based on concrete data with a sample of around 20,000 individuals, their research concludes that these workers experienced "a sense of lost" and "a decline in health on average." According to the

PRC's seventh population census in 2020 (National Bureau of Statistics of China 2020), 28 percent of Fushun's residents are older than 60 years old, and most of them appear to have been affected by the 1990s wave. This situation presumably creates a need to address former workers' sense of loss.

Walking on a safe, comfortable path across the West Open-pit Mine seems to allow these workers to recollect their working memories and reduce their sense of loss to some extent. However, a typical counterargument is that, since not all visitors would respond in the same way as Murakami, some people may feel the opposite when walking through places where negative events have occurred. Though this supposed effect of walking on an industrial site is still in dispute, there is research examining how walking itself can generally relieve emotional distress and improve mental health. For example, Koselka et al. (2019) applied a randomized controlled trial and concluded that positive psychological outcomes are associated with walking. However, few studies directly focus on post-disaster situations, and to what extent Murakami's feelings can generalize to former workers in Fushun remains unclear.

In the stimulus material "How Roads Have Transformed the Natural World," a tension between preservation and redevelopment is discussed. This provides a key insight: green walking path as a type of road may distort original landscape while presumably making the site more accessible (Goldfarb 2023). Besides, since land subsidence will supposedly continue to damage the structures on the ground, the maintenance cost here is likely to be high.

Therefore, the green walking path method appears to have high potential in the "socio-cultural" category, while this potential is not confirmed yet. It probably gets high scores in the environmental and functional categories. Meanwhile, its performance in management and economic categories remains in uncertain.

6. Le Métaphone

6.1. Actions in Le Métaphone

Awarded the 2014 Prix de l'Équerre d'Argent architecture award, the 9-9bis program built a gallery and a music hall based on the remains of a coal factory. Instead of removing the remains and building new modern structures, the program, in collaboration with the architectural firm Hérault Arnod, renewed the 9-9bis mining town through music and art.



Figure 6.1.1 | The Le Metaphone music hall's layout

(<https://www.herault-arnod.fr/projets/9-9bis-le-metaphone/>)

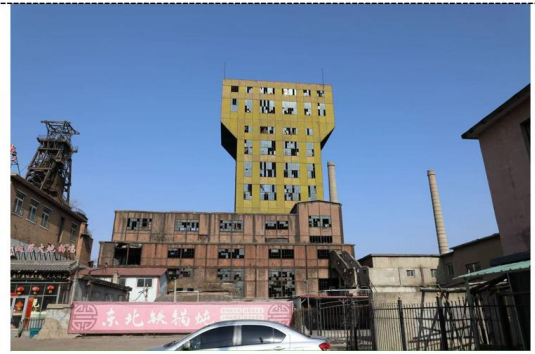
According to the designers' records from Hérault Arnod Architectes, "the idea of music and sound emerged after an analysis of the site's mining past from a social and historical perspective (Hérault Arnod Architectures. n.d.)." This program represents a method of adapting industrial structures into modern buildings for new cultural proposes.

6.2. Counterparts in West Open-pit Mine

The industrial elements of large mining shafts and factory towns appear to present in West Open-pit mine.



A: Longfeng Mining Shaft in 2012



B: Longfeng Mining Shaft in 2026



C: Longfeng left Mining Shaft in 2026

Figure 6.2.1 | Longfeng Mining Shaft (Retrieved from <http://www.fs7000.com/news/?10812.html>)

A: 1st Shale Oil Factory's Remains, shoot in 2026



B: Interior of Shale Oil Factory Remains in 2021



Figure 6.2.2 | 1st Shale Oil' Factory's Remains
(Retrieved from <https://zhuanlan.zhihu.com/p/369560038>)

It appears to support that the West Open-pit mine contains similar yet unutilized industrial elements and therefore provides objects to apply architectural renewal approach.

6.3. Implementation Challenges

First, it seems that not all renewed industrial remains can manage to attract tourists and achieve commercial success. Industrial heritage sites that are far from populated areas and economic centers are likely to face difficulty in attracting tourists. This is shown in Somoza-Medina and Monteserín-Abella's case study (Somoza-Medina and Monteserín-Abella 2021). Their methodology evaluates multiple types of data, including GDP, visitor numbers, and satisfaction reports, which supports their conclusion. Similarly, Fushun's 2023 per capita GDP of 58,468 RMB is lower than the PRC 2023 average of 89,385 RMB and Fushun is located far from China's southern economic center (National Bureau of Statistics of China 2024; Fushun Bureau of Statistics 2024). Therefore, this limitation appears affect Fushun because the city fits the conditions of far from populated economic centers.

Second, remembering history is presumably an important part of preserving industrial heritage. Although the modern approach used by Le Métaphone appears to lead to commercial success, its effectiveness in preserving history remains uncertain. Debary suggests that this approach may cause the public to "forget the harsh realities of industrial decline (Debary 2004)." Despite Debary's scope in Europe probably making his conclusion hard to generalize to Fushun, this still raises a key question that can't be answered yet: When people visit tourism programs because of the entertainment they provide, will they remember the industrial history beneath them?

Therefore, this method appears to have high potential in the architectural category. However, its effect on the economy remains uncertain, and its value in the socio-cultural category is in doubt. The Berlin Wall described in the stimulus material seems to provide an example of how a method that incorporates fashionable elements into a historical site can score highly in socio-cultural category. The Berlin Wall's fragments were presented in various modern styles and attracted artists to work on them, while their effect in reminding humanity of the importance of peace remained significant (Schreiber 2024). However, will West Open-pit mine act in the same way? The answer remains uncertain.

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